

Hyunju Kim

Atlanta, GA, United States

✉ hkim3239@gatech.edu

🌐 hhy0401.github.io

🔗 hhy0401

🔗 Google Scholar

RESEARCH INTERESTS

Computational Neuroscience

Apply deep learning to understand neural dynamics, brain connectivity, and functional representations. Build brain-inspired deep learning models based on neural processing principles.

Graph Data Mining & Graph Algorithms

Design algorithms and AI models for analyzing large-scale, complex graph data.

EDUCATION

Georgia Institute of Technology

Ph.D. Student in Computational Science & Engineering. Advisor: Nabil Imam

Aug. 2024 –

Atlanta, GA, United States

KAIST (Korea Advanced Institute of Science and Technology)

M.S. in Artificial Intelligence. Advisor: Kijung Shin

Feb. 2022 – Feb. 2024

Daejeon, Republic of Korea

POSTECH (Pohang University of Science and Technology)

B.S. in Mathematics. Summa cum laude. GPA 3.96/4.3, Major GPA 4.02/4.3 (ranked 1st)

Feb. 2018 – Feb. 2022

Pohang, Republic of Korea

PUBLICATIONS

*: equal contributions C: Conference J: Journal P: Preprint

- [P2] Geometric Learning for Predicting Functional Connectivity on Folded Cortical Surfaces. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam
- [P1] Geometric Constraints in the Development of Primate Extrastriate Visual Cortex. *Under review*. **Hyunju Kim**, Michael Arcaro, Nabil Imam (doi.org/10.64898/2026.02.04.703881)
- [J3] Four-Set Hypergraphlets for Characterization of Directed Hypergraphs. *TKDE*. Heechan Moon*, **Hyunju Kim***, Sunwoo Kim and Kijung Shin (arxiv.org/abs/2311.14289)
- [J2] Estimating Simplex Counts via Sampling. *The VLDB Journal*. **Hyunju Kim***, Heechan Moon*, Sunwoo Kim and Kijung Shin (DOI: 10.1007/s00778-024-00890-9)
- [J1] Hypergraph Motifs and their Extensions beyond Binary. *The VLDB Journal*. Geon Lee*, Seokbum Yoon*, Jihoon Ko, **Hyunju Kim** and Kijung Shin (DOI: 10.1007/s00778-023-00827-8)
- [C3] DiffIM: Differentiable Influence Minimization with Surrogate Modeling and Continuous Relaxation. *AAAI 2025*. Junghun Lee, **Hyunju Kim**, Fanchen Bu, Jihoon Ko and Kijung Shin
- [C2] FlowerFormer: Empowering Neural Architecture Encoding using a Flow-aware Graph Transformer. *CVPR 2024*. Dongyeong Hwang, **Hyunju Kim**, Sunwoo Kim and Kijung Shin (DOI: 10.1109/CVPR52733.2024.00586)
- [C1] Characterization of Simplicial Complexes by Counting Simplexes beyond Four Nodes. *WWW 2023*. **Hyunju Kim**, Jihoon Ko, Fanchen Bu and Kijung Shin (DOI: 10.1145/3543507.3583332)

RESEARCH EXPERIENCES

EEG Analysis for Cognitive and Clinical Modeling

Sep. 2024 –

Georgia Tech

Drift-Diffusion Model (DDM) analysis of EEG signals for characterizing decision-making processes. EEG-based disease prediction leveraging neural biomarkers and deep learning methods.

fMRI-driven Graph Modeling of Self-Organization Mechanisms in the Visual Cortex

Sep. 2024 –

Georgia Tech

Develop graph generation methods using both AI-driven approaches and rule-based algorithms. Design brain-inspired models by leveraging neural data visualization mechanisms [P1, P2].

Algorithm, Analysis, and Application of Higher-Order Structures

Mar. 2022 – Feb. 2024

KAIST

Designed scalable algorithms for counting and characterizing higher-order structures (simplexes, hypergraphlets, motifs) in simplicial complexes, hypergraphs, and directed hypergraphs. Developed sampling-based estimation methods with provable accuracy guarantees [C1, J1, J2, J3].

KAIST

Developed a null model maintaining performance rankings aligned with the original network while anonymizing users' information.

KAIST

Neural architecture performance prediction method using flow-aware Graph Transformer.

Registered two South Korea Copyrights for Software (C-2022-055775, C-2023-047128).

Patent filed: KR 10-2024-0047575 (filed 2024) [C2].

Supported by IITP grant funded by the Korea government (MSIT).

INDUSTRY EXPERIENCES

Dynamic Stream Triangle Counting Algorithms. Advisor: Kijung Shin

Discrimination of Defective Semiconductors using GNN

Ship Detection: Differentiating True Lines from Noise using CNN

AWARDS & HONORS

PROFESSIONAL SERVICE & TEACHING

Reviewer AAAI 2026, AAAI 2026 (AI Alignment Track), KDD 2026 (AI for Science Track)

SKILLS

Programming Languages C/C++, Python (DL: PyTorch), L^AT_EX, Julia, R, JavaScript
Korean (native), English (fluent: IELTS Overall 7.5; TOEFL – S 24, W 23)

MISCELLANEOUS

POSTECH. IMEN472 Statistical Data Mining term project (1st place, model case). Implemented web crawlers, machine learning algorithms and performed statistical analysis.

POSTECH. UGRP Excellence Award (Advisor: Jaesik Park). Developed and distributed an app platform based on a DL-based model with high accuracy.

POSTECH. Korean I-Corps program for supporting preliminary startups. Title: Development of a Personalized Restaurant and Cafe Recommendation Web Service.

POSTECH. Science Quiz Session. Mathematics representative.

POSTECH. The student council of the Department of Mathematics (Head of Academic Affairs & Public Relations).

Orchestra club 'HanUIRim' (President & Conductor, Feb. 2020 – Dec. 2020).

Amateur Oboist & Classical Music Enthusiast

Guest player of several university orchestras (e.g., Sogang Univ., Sungkyunkwan Univ., Medical School Union, etc.).

Principal oboe player of Yeodo Elementary School Orchestra European tour for promoting International Exposition Yeosu Korea 2012 (Venues: Musikverein, Mozarteum Grosser Saal, Slovak Radio Hall, World Trade Center Patria Hall, Rudolfinum Dvorak Hall, Lukaskirche). 9th Online Supporters, i.e., certified influencers, of Seoul Philharmonic Orchestra for promoting concerts.